

Literature Review: Knowledge Graph-Based Approaches for Fake News Detection

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1 Related Work

Knowledge graph integration has revolutionized fake news detection by enabling factual verification beyond text analysis. Foundational work by [8] introduced DEAP-FAKED, achieving F1-scores of 88% and 78% through NLP and tensor decomposition.

Recent advances employ sophisticated attention mechanisms. [5] utilized graph attention networks for 95% F1-score with concept drift robustness, while [2]’s KAN model implemented knowledge-aware attention for superior multi-dataset performance. [10] enhanced heterogeneous graph neural networks with KG integration, outperforming seven baselines. [7] combined triplet extraction with GenAI embeddings for significant LLM improvements.

Multi-modal approaches have expanded detection capabilities. [6] and [1] integrated fine-grained KGs with multi-modal data through advanced fusion techniques. Cross-domain generalization was addressed by [9] via LLM-inspired heterogeneous graphs and by [3] through multi-domain KG construction.

Interpretability advancements include [4]’s commonsense KG approach achieving 91% accuracy with transparent reasoning. However, computational efficiency and dynamic knowledge integration remain challenging.

Our KNOW-NET architecture addresses these gaps through cross-attention mechanisms and multi-task learning, enabling efficient knowledge integration with minimal data requirements.

References

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